

Exploratory Data Analysis (EDA)

Summary Report

Geldium Financial Delinquency Dataset

1. Introduction

This report details the exploratory analysis of Geldium's customer dataset to assess data quality, identify risk indicators for delinquency, and prepare the dataset for predictive modeling. The goal is to uncover patterns, address data gaps, and provide actionable insights to refine Tata iQ's delinquency risk model.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- Number of records: 500
- Key variables:
 - Demographic: Age, Location, Employment_Status
 - Financial: Income, Credit_Score, Credit_Utilization, Loan_Balance, Debt_to_Income_Ratio
 - Behavioral: Missed_Payments, Month_1–Month_6 payment history, Delinquent_Account (target)
- Data types:
 - Numerical (14): Age, Income, Credit_Score, etc.
 - Categorical (9): Employment_Status, Credit_Card_Type, Location, etc.
 - Binary (1): Delinquent_Account

Anomalies & inconsistencies:

- Credit_Utilization >100%: 3 records (e.g., CUST0090: 102.58%).
- Age-Income mismatch: 18-year-olds with \$199k income; retirees with \$32k.
- Employment_Status: Inconsistent capitalization ("EMP", "employed", "retired").

- Payment history: Spelling variations ("On time" vs. "On-time").

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- Variables with missing values:

Column	Missing Count	Handling Method	Justification
Income	14	Median imputation by Employment_Status	Preserves distribution while accounting for income differences across job types.
Loan_Balance	12	Median imputation by Credit_Card_Type	Reflects typical balances per card tier (e.g., Platinum vs. Student).
Credit_Score	1	Mean imputation by Age group	Credit scores correlate strongly with age brackets.

Additional treatments:

- Credit_Utilization: Capped at 100% for 3 records to align with industry standards.
- Debt_to_Income_Ratio: Scaled values >1 down by 100 (e.g., 0.1 → 10%).
- Categorical standardization: Lowercased Employment_Status and standardized payment history labels.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Correlations and risk patterns:

1. Strong delinquency predictors:
 - Missed_Payments (≥4): 68% of delinquent accounts had ≥4 missed payments (vs. 12% for non-delinquent).
 - Credit_Utilization >70%: 78% of high-utilization customers had ≥3 late/missed payments in 6 months.

- Sequential poor payment history: Patterns like "Late → Missed → Late" in Month_1-6 predicted 89% of delinquencies.
2. Unexpected anomalies:
 - High-income delinquents: 7% of delinquent customers earned >\$150K/year (e.g., CUST0002: \$101K, Self-employed, 6 missed payments).
 - Retiree risk: 11 retirees had delinquency flags, potentially linked to fixed-income constraints (e.g., CUST0059: \$81K income, 4 missed payments).
 - Location bias: Phoenix customers had 22% higher delinquency rates than New York.
 3. Demographic insights:
 - Age: 18–25 year-olds showed 37% higher delinquency risk.
 - Employment: Unemployed customers were 2.1× more likely to be delinquent than employed.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

GenAI prompts and outputs:

- Prompt: "Identify top 3 delinquency predictors in this dataset with reasoning."
 - Output: Missed_Payments (direct behavior), Credit_Utilization (liquidity stress), Payment History sequences (trend consistency).
- Prompt: "Suggest imputation for missing Income values using industry best practices."
 - Output: Median by Employment_Status—avoids skew from income outliers.
- Prompt: "Detect anomalies in Age-Income relationships."
 - Output: Flagged teens with \$199K income and retirees below poverty line.

6. Conclusion & Next Steps

Summary:

The dataset reveals strong delinquency signals in payment behavior and credit utilization, but requires cleaning for missing values and anomalies. Key risks include youth inexperience, unemployment, and geographic disparities.

Next steps:

- Feature engineering: Create a Payment_Stability_Score from Month_1–6 sequences.

- Bias mitigation: Audit model for location/age discrimination using SHAP values.
- Model prioritization: Focus on Logistic Regression and Random Forests for interpretability.
- Data expansion: Collect additional variables (e.g., loan purpose, dependents) to explain high-income delinquencies.

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